Plasmid Cloning

Introduction

Protocol is based on a FastDigest digestion protocol and an "in-house" cloning protocol.

This is for CROP-seq opti plasmid.

Good protocol for gel electrophoresis.

Materials

- Plasmid of interest
- Qubit reagents (1x BR)
  - Tubes, buffer, and standards
- 10X FastDigest Buffer
- FastDigest enzyme
- DPEC water
- Agarose powder
- TBE
- Ladder (master mix or make self with 10X BlueJuice and DPEC water)
- 6X loading buffer
- Oligos/primers
- 10X T4 Ligation Buffer (NEB)
- T4 PNK (NEB)
- 2X Quickligation Buffer (NEB)
- Quick Ligase (NEB)
- Gel extraction kit
- Equipment
  - Qubit
  - Owl gel electrophoresis
  - Imager
  - PCR mahcine

Procedure

Digestion

1. Determine concentration of plasmid via Qubit. Calculate volume needed for 1 ug.

2. Digest 1 ug plasmid with BsmB1/Esp3I (restriction enzyme) for 30 minutes at 37C.

    Add:
    13 uL DPEC water
    2 uL 10X FastDigest Buffer
    1 uL DTT (20 mM)
    ___ uL plasmid (1 ug) (3 uL in this case)
    1 uL FastDigest enzyme
    20 uL total

Purification and Extraction

3. Setup gel electrophoresis to purify digested plasmid.

    Mix a 1% agarose gel: 50 mL TBE to 500 mg agarose powder in a glass beaker
    Microwave for 20 second intervals and swirl to mix. Do so for about < 1 min, or until all the powder is dissolved
    Let agarose solution cool down to about 50 °C (about when you can comfortably keep your hand on the flask), about 5 mins
    During this time, set up the Owl electrophoresis system: tighten and add comb (~ 6 prongs, directly depends on amount of samples)
    Add 5 uL Sybersafe to gel once slighlty cooled and add to machine
    Let harden (about 10 min)

4. Load ladder and samples.

    Once solidified, place the agarose gel within its holder into the electrophoresis unit
    Fill up to max line with TBE
    Load ladder master mix (ladder + 10X BlueJuice + water)
    Load digested plasmid (20 uL) from previous step
    Load control: add 2.5 uL of undigested plasmid to 0.5 uL of 6X loading buffer

5. Cover, plug in, turn on, and run gel at 100 V for an hour.

6. Take an image of the gel and identify target band. Use a disposible blue razor thing to remove your target band and place in eppendorf tube.

7. Run gel extraction kit and elute in EB

    link kit!! whcih one??

8. Qubit to find concentration.

Phosphorylate and Anneal

9. Set up PCR machine to run at below incubation in step 10.

10. Add to a PCR tube:

    1 uL oligo 1 (100 uM)

    1 uL oligo 2 (100 uM)

    1 uL 10X T4 Ligation Buffer (NEB)

    6.5 uL DPEC water

    0.5 uL T4 PNK (NEB)

    10 uL total

11. Incubate! And store 4C.

    37C for 30 min

    95C for 5 min

    Ramp down to 25C at 5C/min

Ligate

12. Dilute phosphorylated and annealed oligo duplex from previous step 1:200

    1 uL duplex + 199 uL DPEC water

13. Add:

    __uL purified and digested plasmid from step 7 (50 ug).

    1 uL diluted oligo duplex from previous step

    5 uL 2X Quickligation Buffer (NEB)

    __uL DPEC water (up to 10 uL total)

    10 uL subtotal

    1 uL Quick Ligase (NEB)

    11 uL total

Continue to Bacterial Transformation